The documentation and process conversion measures necessary to comply with this revision shall be completed by 21 Mar 94.

INCH-POUND

MIL-S-19500/533D
20 September 1993
SUPERSEDING
MIL-S-19500/533C
16 December 1991

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, ZENER, VOLTAGE REGULATOR, SOLID GLASS NONCAVITY CONSTRUCTION, TYPES 1N6309 THROUGH 1N6355; 1N6309US THROUGH 1N6336US; PLUS C AND D TOLERANCE SUFFIX; JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for microminiature 500 mW, silicon, metallurgically bonded, voltage regulator diodes with tolerances of 5 percent, 2 percent, and 1 percent. Four levels of product assurance are provided for each device as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (DO-35) and figure 2 (surface mount).

1.3 Maximum ratings. Maximum ratings are as shown in columns 6, 7, and 9 of table IV herein, and as follows:

P_T = 500 mW, surface mount device with T_{EC} = +125°C.
P_T = 500 mW at T_L = +75°C, L = .375 inch (9.53 mm); both ends of case or diode body to heat sink at L = .375 inch (9.53 mm). (Derate I_Z to 0.0 mA dc at +175°C.)
-65°C ≤ T_{op} ≤ +175°C; -65°C ≤ T_{STG} ≤ +175°C.

1.4 Primary electrical characteristics at T_A = +25°C. Primary electrical characteristics are shown in columns 2, 3, and 11 of table IV herein, and as follows:

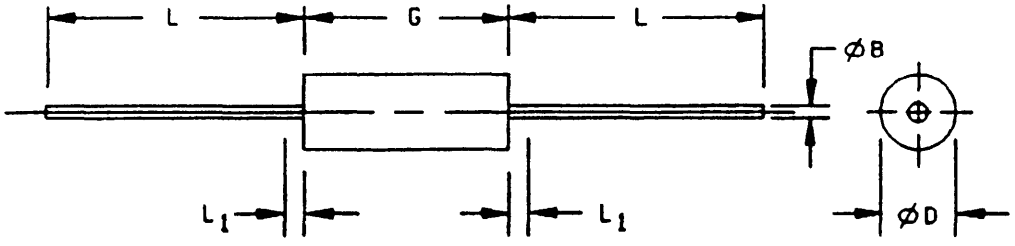
2.4 V dc ≤ V_Z ≤ 200 V dc (nominal).
1N6309D through 1N6355D are 1 percent voltage tolerance.
1N6309C through 1N6355C are 2 percent voltage tolerance.
1N6309 through 1N6355 are 5 percent voltage tolerance.

$R_{\theta JL}$ = 250°C/W (maximum) at L = .375 inch (9.53 mm) nonsurface mount.
$R_{\theta JL}$ = 50°C/W (maximum) at L = 0 inch nonsurface mount.
$R_{\theta JEC}$ (end cap) = 50°C/W (maximum) surface mount.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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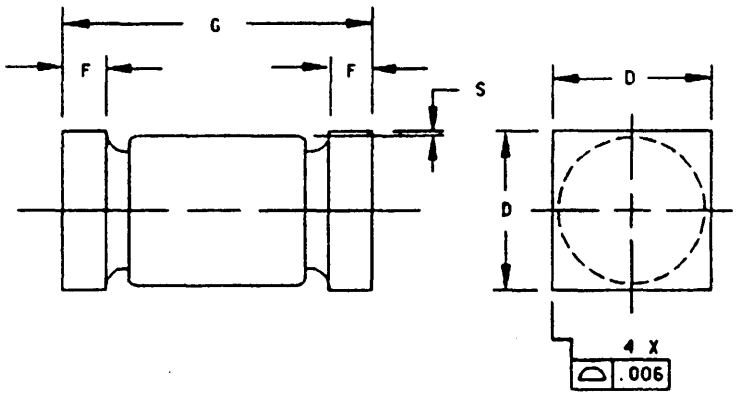


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ØB	.018	.022	.46	.56	
ØD	.060	.090	1.52	2.29	
G	.120	.200	3.05	5.08	
L	1.000	1.500	25.40	38.10	
L_1		.050		1.27	3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Lead diameter not controlled in this zone to allow for flash, lead finish build-up, and minor irregularities other than slugs.

FIGURE 1. Physical dimensions, (DO-35) nonsurface mount devices.

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Symbol	D-5D			
	Inches		Millimeters	
	Min	Max	Min	Max
G	.165	.195	4.19	4.95
F	.019	.028	0.48	0.71
S	.003		0.08	
D	.070	.096	1.78	2.16

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions, surface mount devices.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows:

EC- - - - - - - - - - - - End cap.

T_{EC} - - - - - - - - - - - - Temperature, end cap.

US- - - - - - - - - - - - Surface mount, square end cap.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figures 1 and 2 herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500 and MIL-STD-750. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 Diode construction. All devices shall be metallurgically bonded, thermally matched, noncavity, double plug construction in accordance with MIL-S-19500, appendix A, and 3.3.3.1 and 3.3.3.2 herein.

3.3.2.1 Metallurgical bond for diodes with $V_Z \geq 6.8$ V dc. Category I bonds as defined in accordance with MIL-S-19500 shall be utilized.

3.3.2.2 Metallurgical bond for diodes with $V_Z \leq 6.8$ V dc. Category I, category II, or category III bonds as defined in accordance with MIL-S-19500 may be utilized.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

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3.4.1 Marking of US version. For US version only; all marking may be omitted from the device except for the cathode marking. All marking which is omitted from the body of the device shall appear on the label of the initial container.

3.4.2 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end. Alternately for surface mount (US) devices, a minimum of three evenly spaced contrasting color dots around the periphery of the cathode end may be used. No color coding will be permitted.

3.5 Selection of tight tolerance devices. The C and D suffix devices shall be selected from JAN, JANTX, JANTXV, or JANS devices, which have successfully completed all applicable screening, and groups A, B, and C testing as 5 percent tolerance devices. All sublots of C and D suffix devices shall pass group A, subgroup 2, at tightened tolerances. Tighter tolerances for mounting clip temperature shall be maintained for reference purpose to establish correlation. For C and D tolerance levels, T_L = 30 ± 2°C at 0.375" from body or equivalent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein. Lot accumulation period shall be 6 months in lieu of 6 weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-S-19500 and table II herein.

4.3 Screening (JANS, JANTX, AND JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3a	Temperature cycling	Temperature cycling
3c 1/	Thermal impedance (see 4.5.7)	Thermal impedance (see 4.5.7)
9	I_{R1}, V_{Z2}, and Z_Z	Not applicable
11	I_{R1} and V_{Z2}; ΔI_{R1} ≤ 100 percent of initial value or 50 nA dc, whichever is greater. ΔV_{Z2} ≤ ±1 percent of initial value; ΔZ_Z ≤ ±35 percent of initial value or 2Ω, whichever is greater.	I_{R1} and V_{Z2}
12	See 4.2.1	See 4.2.1, t = 96 hours
13(a) 2/	Subgroups 2 and 3 of table I herein; ΔI_{R1} ≤ 100 percent of initial value or 50 nA dc, whichever is greater; ΔV_{Z2} ≤ ±1 percent of initial value.	Subgroup 2 of table I herein; ΔI_{R1} ≤ 100 percent of initial value or 50 nA dc, whichever is greater; ΔV_{Z2} ≤ ±1 percent of initial value.

1/ Thermal impedance may be performed anytime after sealing provided temperature cycling is performed in accordance with MIL-S-19500, screen 3 prior to this test.

2/ $Z_{\theta JX}$ need not to be performed at screen 13, if performed prior to screen 13.

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4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

I_Z = column 7 of table IV; mounting conditions in accordance with method 1026 of MIL-STD-750;
T_A = room ambient as defined in 4.5 of MIL-STD-750;
T_{EC} = +100°C to +125°C for surface mount devices.

4.3.2 Scope display evaluation. The reverse breakdown region of the voltage-current characteristic shall be viewed over the applicable voltage-current range (see table IV, columns 2 and 3) on an oscilloscope with a presentation magnitude of at least 5 cm in both the horizontal and vertical direction. Reverse peak power over the knee shall be at least 10 mW pk. Any discontinuity or dynamic instability of the trace shall be cause for rejection of the device, as described in MIL-STD-750, method 4023.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. End-point electrical measurements shall be in accordance with the applicable steps of table III herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection: $Z_{\theta JX} \leq 15°C/W$.

I_H = 2 A minimum.

t_H = 10 ms.

I_M = 1 mA to 10 mA.

t_{MD} = 100 µs maximum.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
3	4066	T_A = +25°C ±5°C (see 4.5.1); I_{ZSM} = column 9 of table IV (shall be performed on each sublot).
4	1037	I_Z = column 7 of table IV at T_A = room ambient as defined in 4.5 of MIL-STD-750; $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles. Mounting conditions in accordance with 4.2.1. No forced air cooling on the device shall be permitted.
5	1027	T_A = +125°C ±25°C for 96 hours; I_Z(min) = column 7 of table IV with I_Z adjusted as required by the chosen T_A to give an average lot T_J = +275°C. Marking legibility requirements shall not apply. Mounting conditions in accordance with 4.2.1.
6	3101 or 4081	+25°C ≤ T_L ≤ +35°C; see figure 4.

4.4.2.2 Group B inspection, table IVb (JAN, JANTX, and JANTXV of MIL-S-19500).

Subgroup	Method	Condition
2	4066	T_A = +25°C ±5°C (see 4.5.1); I_{ZSM} = column 9 of table IV (shall be performed on each sublot).
4	1027	I_Z(min) = 50 percent of column 7 of table IV (mounting conditions in accordance with 4.2.1); T_A = room ambient as defined in 4.5 of MIL-STD-750.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Condition
2	2036	Test condition A, 4 pounds, t = 15 s ± 3 s Test condition E, 8 ounces, t = 15 s ± 3 s not applicable to surface mount devices.
6	1026	I_Z(min) = 50 percent of column 7 of table IV (mounting conditions in accordance with 4.2.1); T_A = room ambient as defined in 4.5 of MIL-STD-750.
7	4071	(For JAN, JANTX, and JANTXV only); I_Z = column 4 of table IV; T_1 = +25°C ±5°C; T_2 = +125°C (see 4.5.4); LTPD = 10.
8	1056	20 cycles of thermal shock from Liquid N_2 (-195°C) to +150°C fluoro bath. The diode under test (DUT) shall be stabilized at the temperature extremes for 20 seconds (minimum); transfer time, ≤ 5 s continuously monitor during the last five cycles V_f at 500 mA pk (with an open-circuit voltage clamped at 3 V maximum) on Tektronix 576 scope or equivalent throughout temperature excursions and at temperature extremes for any discontinuity instability hysteresis or other anomaly shall cause for rejection of the device. 20 cycles for group C. LTPD = 10.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Surge current (I_{ZSM}). The peak currents specified in column 9 of table IV shall be applied in the reverse direction and these shall be superimposed on the current I_{Z2} (table IV, column 4); a total of five surges at one-minute intervals. Each individual surge shall be .5-square-wave-pulse of .08-second duration or an equivalent sine wave with the same effective rms current.

4.5.2 Regulator voltage measurements. The test current shall be applied until thermal equilibrium is attained (90 seconds minimum) prior to reading the breakdown voltage. For this test, the diode shall be suspended by its leads with mounting clips whose inside edge is located at .375 inch (9.53 mm) from the body and the mounting clips shall be maintained at a temperature of +25°C +8°C, -2°C. US suffix devices shall be mounted at the end caps. This measurement may be performed after a shorter time following application of the test current than that which provides thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the Government.

4.5.3 Voltage regulation. For values of V_{Z2} (nominal) from 6.8 V dc to 200 V dc, current at 10 percent of I_{ZM} (column 7) shall be maintained for a period of 90 ±5 seconds, and then V_Z shall be recorded. The current shall then be increased to 50 percent of I_{ZM} (column 7) and maintained for a period of 90 ±5 seconds, and then V_Z shall be recorded. The voltage change shall not exceed the applicable limits as shown in table IV, column 8. During this test, the diode shall be suspended by its leads with mounting clips whose inside edge is located between .375 inch (9.53 mm) and .500 inch (12.70 mm) from the device body and the mounting clips shall be maintained at a temperature of +25°C +8°C, -2°C. US suffix devices shall be mounted by the end caps. For values of V_{Z2} (nominal) from 2.4 V dc to 6.2 V dc, the lower test current shall be 2 mA dc and the higher current shall be 20 mA dc.

4.5.4 Temperature coefficient of regulator voltage (α_{VZ}). The device shall be temperature stabilized with current applied prior to reading regulator voltage at the specified ambient temperature.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits 2/		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	(See 4.4.1)	$Z_{\theta JX}$		15	°C/W
Forward voltage	4011	I_F = 1 A dc, pulsed (see 4.5.6)	V_F		1.4	V dc
Reverse current	4016	DC method, V_R = column 10 of table IV	I_{R1}		column 11	µA dc
Regulator voltage	4022	I_{Z1} = 250 µA dc (see 4.5.2)	V_{Z1}	column 3		V dc
Regulator voltage	4022	I_{Z2} = column 4 of table IV (see 4.5.2)	V_{Z2}	column 2 -5, -2, -1 percent	column 2 +5, +2, +1 percent	V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
Reverse current	4016	DC method, V_R = column 10 of table IV	I_{R2}		column 12	µA dc
Subgroup 4						
Small-signal reverse breakdown impedance	4051	I_Z = column 4 of table IV, I_{sig} = 10 percent of I_{Z2}	Z_Z		column 5	Ω
Knee impedance	4051	I_{ZK} = 250 µA dc, I_{sig} = 25 µA rms	Z_{ZK}		column 6	Ω
Noise density		I_{Z1} = 250 µA dc (see 4.5.5)	N_D		column 14	$\mu V/\sqrt{Hz}$
Capacitance		V_R = 0 V dc, 100 kHz ≤ f ≤ 1 MHz	C		column 16	pf
Subgroup 5 and 6						
Not applicable						

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits 2/		Unit
	Method	Conditions		Min	Max	
Subgroup 7		116 devices, c = 0				
Scope display evaluation		See 4.2.2				
Subgroup 8		LTPD = 10				
Voltage regulation		(See 4.5.3)	$V_{Z(reg)}$		column 8	V dc
Temperature coefficient of regulator voltage	4071	JANS level, IZ = column 4 of table IV, T_1 = +25°C ±5°C; T_2 = +125°C (see 4.5.4)	α_{VZ}		column 15	%/°C

1/ For sampling plan, see MIL-S-19500.
2/ Column references are to table IV.

TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Sampling plan 1/	Small lot quality conformance inspection n/C
	Method	Conditions		
Subgroup 1			5	12/0
Temperature cycling	1051	500 cycles, condition C		
Electrical measurement		See table III, steps 2 and 3		
Subgroup 2				
Not applicable				
Subgroup 3				
Not applicable				
Subgroup 4			5	12/0
Thermal resistance	3101 or 4081	(See 4.5.8)		

1/ For sampling plans not specified, see MIL-S-19500.

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TABLE III. Groups B, C, and E electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits 4/		Unit
		Method	Conditions		Min	Max	
1	Reverse current	4016	DC method, V_R = column 10 of table IV	I_{R1}		column 11	µA dc
2	Reverse current	4016	DC method, V_R = column 10 of table IV	I_{R3}		column 13	µA dc
3	Forward voltage	4011	I_F = 1 A dc	V_F		1.4	V dc
4	Regulator voltage	4022	I_Z = column 4 of table IV (See 4.5.2)	V_{Z2}	column 2 -5 percent	column 2 +5 percent	V dc
5	Knee impedance	4051	I_{ZK} = 250 µA dc, I_{sig} = 25 µA rms	Z_{ZK}		column 6	Ω
6	Small-signal breakdown impedance	4051	I_Z = column 4 of table IV, I_{sig} = 10 percent of I_Z	Z_Z		column 5	Ω
7	Regulator voltage	4022	I_Z = column 4 of table IV (See 4.5.2)	ΔV_{Z2}		±1 percent of initial value.	
8	Reverse current	4016	DC method, V_R = column 10 of table IV	ΔI_{R1}		100 percent of initial value or 50 nA dc, whichever is greater.	
9	Small-signal breakdown impedance	4051	I_Z = column 4 of table IV, I_{sig} = 10 percent of I_Z	ΔZ_Z		±35 percent of initial value or 2 Ω, whichever is greater.	

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table III herein, steps 1, 3, 4, and 5.

b. Subgroup 4, see table III herein, steps 1, 3, 4, 5, 6, 7, 8, and 9.

c. Subgroup 5, see table III herein, steps 2, 3, 4, 5, 6, 7, 8, and 9.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 3, and 4.

b. Subgroup 3, see table III herein, steps 2, 3, 4, and 7.

c. Subgroup 6, see table III herein, steps 2, 3, and 4.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroups 2 and 3, see table III herein, steps 1, 3, 4, and 5 for JANS and steps 1, 3, and 4 for JAN, JANTX, and JANTXV.

b. Subgroup 6, see table III herein, steps 1, 3, 4, and 5 for JANS and steps 2, 3, and 4 for JAN, JANTX, and JANTXV.

4/ Column references are to table IV.

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TABLE IV. Characteristics and ratings.

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 11	Col 12	Col 13	Col 14	Col 15	Col 16
Type	V_{Z2} nom at I_{Z2} 1/	V_{Z1} min at I_{Z1} 250 µA	I_{Z2} test cur-rent	Z_Z at I_{Z2}	Z_{ZK} at 250 µA	I_{ZM}	$V_{Z(reg)}$	I_{ZSM} surge	V_R	I_{R1} at +25°C	I_{R2} at T_A =+150°C	I_{R3} life	N_D at 250 µA 1-3 kHz	α_{VZ}	C at 0 V
	V	V	mA	Ω	Ω	mA	V	A	V	µA	µA	µA	µV/√Hz	%/°C	pF
1N6309	2.4	1.1	20	30	1,200	177	1.50	2.50	1.0	100	200	150	1.0	-.085	2,000
1N6310	2.7	1.2	20	30	1,300	157	1.50	2.20	1.0	60	150	100	1.0	-.080	1,900
1N6311	3.0	1.3	20	29	1,400	141	1.50	2.00	1.0	30	100	60	1.0	-.075	1,800
1N6312	3.3	1.5	20	24	1,400	128	1.60	1.80	1.0	5	20	10	1.0	-.070	1,650
1N6313	3.6	1.8	20	22	1,400	117	1.60	1.65	1.0	3	12	6	1.0	-.065	1,600
1N6314	3.9	2.0	20	20	1,700	108	1.60	1.50	1.0	2	12	4	1.0	-.060	1,400
1N6315	4.3	2.4	20	18	1,700	99	0.90	1.40	1.0	2	12	4	1.0	-.045 +.020	1,350
1N6316	4.7	2.8	20	16	1,500	90	0.50	1.27	1.5	5	12	10	1.0	-.028 +.032	1,300
1N6317	5.1	3.3	20	14	1,300	83	0.40	1.17	2.0	5	12	10	1.0	-.020 +.035	1,200
1N6318	5.6	4.3	20	8	1,200	76	0.40	1.10	2.5	5	10	10	2.0	+.050	1,150
1N6319	6.2	5.2	20	3	800	68	0.30	0.97	3.5	5	10	10	5.0	+.060	1,050
1N6320	6.8	6.0	20	3	400	63	0.35	1.23	4.0	2	50	4	5.0	+.062	1,000
1N6321	7.5	6.6	20	4	400	57	0.40	1.16	5.0	2	30	4	5.0	+.068	900
1N6322	8.2	7.5	20	5	400	52	0.40	1.07	6.0	1	10	2	20.0	+.075	800

See footnotes at end of table.

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TABLE IV. Characteristics and ratings - Continued.

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 11	Col 12	Col 13	Col 14	Col 15	Col 16
Type	V_{Z2} nom at I_{Z2} 1/	V_{Z1} min at I_{Z1} 250 µA	I_{Z2} test cur-rent	Z_Z at I_{Z2}	Z_{ZK} at 250 µA	I_{ZM}	$V_{Z(reg)}$	I_{ZSM} surge	V_R	I_{R1} at +25°C	I_{R2} at T_A =+150°C	I_{R3} life	N_D at 250 µA 1-3 kHz	α_{VZ}	C at 0 V
	V	V	mA	Ω	Ω	mA	V	A	V	µA	µA	µA	$\mu V/\sqrt{Hz}$	%/°C	pF
1N6323	9.1	8.4	20.0	6	500	47	0.50	0.97	7.0	1.00	10	2.000	40	+.076	700
1N6324	10.0	9.1	20.0	6	500	43	0.50	0.89	8.0	1.00	10	2.000	80	+.079	600
1N6325	11.0	10.0	20.0	7	550	39	0.50	0.83	8.5	1.00	10	2.000	100	+.082	500
1N6326	12.0	11.0	20.0	7	550	35	0.55	0.77	9.0	1.00	10	2.000	100	+.083	450
1N6327	13.0	11.9	9.5	8	550	33	0.55	0.71	9.9	0.05	10	0.100	100	+.079	400
1N6328	15.0	13.8	8.5	10	600	28	0.70	0.62	11.0	0.05	10	0.100	100	+.082	350
1N6329	16.0	14.7	7.8	12	600	27	0.75	0.58	12.0	0.05	10	0.100	100	+.083	325
1N6330	18.0	16.6	7.0	14	600	24	0.85	0.52	14.0	0.05	10	0.100	100	+.085	300
1N6331	20.0	18.5	6.2	18	500	21	0.95	0.47	15.0	0.05	10	0.100	100	+.086	275
1N6332	22.0	20.4	5.6	20	500	19	1.05	0.43	17.0	0.05	10	0.100	100	+.087	260
1N6333	24.0	22.3	5.2	24	500	18	1.15	0.39	18.0	0.05	10	0.100	100	+.088	240
1N6334	27.0	25.2	4.6	27	500	16	1.30	0.35	21.0	0.05	10	0.100	100	+.090	220
1N6335	30.0	28.0	4.2	32	500	14	1.45	0.31	23.0	0.05	10	0.100	100	+.091	200
1N6336	33.0	30.9	3.8	40	600	13	1.60	0.28	25.0	0.05	10	0.100	100	+.092	185

See footnotes at end of table.

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TABLE IV. Characteristics and ratings - Continued.

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 11	Col 12	Col 13	Col 14	Col 15	Col 16
Type	V_{Z2} nom at I_{Z2} 1/	V_{Z1} min at I_{Z1} 250 µA	I_{Z2} test current	Z_Z at I_{Z2}	Z_{ZK} at 250 µA	I_{ZM}	$V_{Z(reg)}$	I_{ZSM} surge	V_R	I_{R1} at +25°C	I_{R2} at T_A =+150°C	I_{R3} life	N_D at 250 µA 1-3 kHz	α_{VZ}	C at 0 V
	V	V	mA	Ω	Ω	mA	V	A	V	µA	µA	µA	$\mu V/\sqrt{Hz}$	%/°C	pF
1N6337	36.0	33.7	3.4	50	600	12.0	1.75	0.260	27	0.05	10	0.100	100	+.093	175
1N6338	39.0	36.6	3.2	55	700	11.0	1.90	0.240	30	0.05	10	0.100	100	+.094	170
1N6339	43.0	40.4	3.0	65	800	9.9	2.10	0.220	33	0.05	10	0.100	80	+.095	165
1N6340	47.0	44.2	2.7	75	900	9.0	2.25	0.200	36	0.05	10	0.100	80	+.095	155
1N6341	51.0	48.0	2.5	85	1,000	8.3	2.50	0.180	39	0.05	10	0.100	80	+.096	145
1N6342	56.0	52.7	2.2	100	1,200	7.6	2.70	0.170	43	0.05	10	0.100	80	+.097	135
1N6343	62.0	58.4	2.0	125	1,300	6.8	2.90	0.150	47	0.05	10	0.100	80	+.097	130
1N6344	68.0	64.1	1.8	155	1,500	6.3	3.20	0.130	52	0.05	10	0.100	80	+.098	120
1N6345	75.0	70.8	1.7	180	1,600	5.7	3.40	0.125	56	0.05	10	0.100	80	+.098	110
1N6346	82.0	77.4	1.5	220	1,800	5.2	3.80	0.115	62	0.05	10	0.100	80	+.099	105
1N6347	91.0	86.0	1.4	270	2,100	4.7	4.20	0.100	69	0.05	10	0.100	80	+.099	100
1N6348	100.0	94.5	1.3	340	2,400	4.3	4.40	0.095	76	0.05	10	0.100	80	+.110	95
1N6349	110.0	104.0	1.1	500	2,800	3.9	4.80	0.085	84	0.05	10	0.100	80	+.110	90
1N6350	120.0	113.0	1.0	600	3,200	3.5	5.20	0.080	91	0.05	10	0.100	80	+.110	70

See footnotes at end of table.

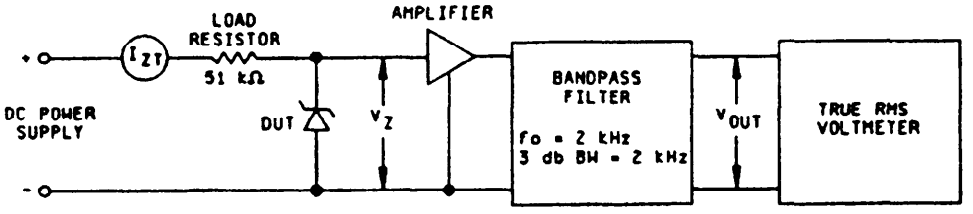
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TABLE IV. Characteristics and ratings - Continued.

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 11	Col 12	Col 13	Col 14	Col 15	Col 16
Type	V_{Z2} nom at I_{Z2} 1/	V_{Z1} min at I_{Z1} 250 µA	I_{Z2} test current	Z_Z at I_{Z2}	Z_{ZK} at 250 µA	I_{ZM}	$V_{Z(reg)}$	I_{ZSM} surge	V_R	I_{R1} at +25°C	I_{R2} at T_A =150°C	I_{R3} life	N_D at 250 µA 1-3 kHz	α_{VZ}	C at 0 V
	V	V	mA	Ω	Ω	mA	V	A	V	µA	µA	µA	$\mu V/\sqrt{Hz}$	%/°C	pF
1N6351	130.0	122	0.95	850	4,100	3.3	5.60	0.070	99	0.05	10	0.100	80	+.110	70
1N6352	150.0	141	0.85	1,000	4,500	2.8	7.00	0.065	114	0.05	10	0.100	80	+.110	65
1N6353	160.0	151	0.80	1,200	5,000	2.7	7.50	0.060	122	0.05	10	0.100	80	+.110	65
1N6354	180.0	170	0.68	1,500	5,600	2.4	9.00	0.050	137	0.05	10	0.100	80	+.110	60
1N6355	200.0	189	0.65	1,800	6,500	2.1	12.00	0.045	152	0.05	10	0.100	80	+.110	55

1/ 1N6309D through 1N6355D are 1 percent voltage tolerance.
1N6309C through 1N6355C are 2 percent voltage tolerance.
1N6309 through 1N6355 are 5 percent voltage tolerance.

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NOTES:

1. Input voltage and load resistance should be high so that the DUT can be driven from a constant current source.
2. Input impedance of band pass filter should be high compared with the dynamic impedance of the DUT.
3. Filter bandwidth characteristics shall be as follows:

$$f_o = 2{,}000 \text{ Hz}$$

Shape factor, -40 dB to -3 dB, approximately 2.
Passband at the -3 dB point is 1,000 Hz ±50 Hz to 3,000 Hz ±150 Hz.
Passband at the -40 dB point is 500 Hz ±50 Hz to 6,000 Hz ±600 Hz.

FIGURE 3. Circuit for determination of noise density.

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4.5.5 Noise density. Noise voltage shall be measured using a noise density test circuit as shown on figure 3. Place a low-noise resistor, equivalent in value to the dynamic impedance of the DUT, in the test clips and adjust test current (I_{ZT}) to 250 µA dc and measure output-noise voltage. Remove the resistor, insert DUT in test clips, readjust test current to 250 µA dc and measure output-noise voltage again. To obtain noise density (N_D), subtract rms resistor output-noise voltage from rms diode output-noise voltage and divide by product of overall system gain and square root of bandwidth. All measurements shall be made at T_A = +25°C.

4.5.6 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.7 Thermal impedance. Thermal impedance $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit for $Z_{\theta JX}$ in screening (table II of MIL-S-19500) shall not exceed the group A, subgroup 2 limit). (For conditions, see 4.4.1.)

4.5.7.1 For initial qualification and requalification. Read and record data ($Z_{\theta JX}$) shall be supplied to the qualifying activity on one lot (random sample of 500 devices minimum) prior to shipment. Twenty-two samples shall be serialized and provided to the qualifying activity for test correlation.

4.5.8 Thermal resistance. Thermal resistance measurement shall be performed in accordance with MIL-STD-750, method 3101 or 4081. Read and record data in accordance with group E herein and shall be included in the qualification report. Forced moving air or draft shall not be permitted across the devices during test. The maximum limit for $R_{\theta JL}$ under these condition shall be $R_{\theta JL(max)}$ = 250°C/W, for surface mount $R_{\theta JEC}$ = 50°C/W. The following conditions shall apply:

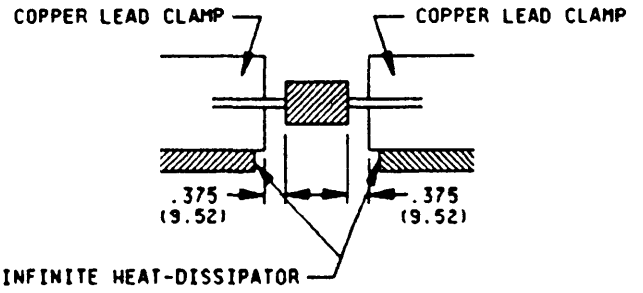
I_H = 200 mA to 400 mA.

t_H = 25 seconds minimum.

I_M = 1 mA to 10 mA.

t_{MD} = 100 µs maximum.

LS = Lead spacing = .375 inch (9.53 mm) as defined on figure 4.



NOTES:
1. The lead temperature, T_L shall be measured on a lead at a point adjacent to the heat sink clamp (reference point).
2. The clamping force on each lead shall be 4 ±0.5 pounds.
3. The DUT shall be shielded from drafts.
4. The heat sink clamps shall be placed equal distances from each end of the diode body.
5. For surface mount devices, the end caps shall be clamped to the heat sinks.

FIGURE 4. Mounting details for thermal resistance.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Type designation and quality assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Navy - TD
Air Force - 19, 85
DLA - ES

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5961-1527)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/533D	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, DIODE, SILICON, ZENER, VOLTAGE REGULATOR, SOLID GLASS, NONCAVITY CONSTRUCTION, TYPES 1N6309 THROUGH 1N6355; 1N6309US THROUGH 1N6336US; PLUS C AND D TOLERANCE SUFFIX; JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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